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International AS

Physics (9630)
PH01 Mechanics, materials and atoms
Report on the Examination

June 2025

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REPORT ON THE EXAMINATION: INTERNATIONAL AS PHYSICS (9630) PH01 – JUNE 2025

This paper was similar to those of previous years in terms of specification coverage and demand. Students had many opportunities to demonstrate their knowledge, skills and understanding across a range of topics. The usual variety of question styles including short answers, single and multi-step calculations, extended writing, and multiple choice were used. The questions were set in a range of both familiar and unfamiliar contexts.
The balance of assessment objectives and the mathematical demand was similar to those of previous series.

QUESTION 01

This question required students to describe the behaviour of different materials. These questions were context-free, so should have been quite straightforward for students. Some students wrote incomplete answers that did not give an accurate enough definition of the behaviours.

Nearly 70% of students were able to describe plastic behaviour in question 01.1. Some students wrote vague answers that stated that the material would change length or shape but did not state that the change was permanent.

In question 01.2 only 30% of students were able to describe the behaviour of a brittle material. Many students who did not gain the mark did state that the material did not undergo plastic deformation but did not give a complete answer and state that there is no plastic deformation before failure.

QUESTION 02

Question 2 required students to state the conditions necessary for the two forces to form a couple. The question was reasonably accessible, with a mean mark of 61%. Most students were able to give one of the conditions but only 40% could give two. Most of the students who only gave one condition stated that the forces are equal in magnitude and opposite in direction but did not state that they are separated by a perpendicular distance.

QUESTION 03

This question was about a conical spring, so was an unfamiliar context. The first part of the question was well answered, but some students found interpreting the graph challenging and were unable to answer the last two parts of the question well.

For question 03.1 approximately 70% of students were able to determine the fundamental (base) units of the stiffness of a spring. Some students simply stated that the units were N/m.

In question 03.2 40% of students failed to realise that the area under the curve in figure 3 was equal to the work done compressing the spring, so were unable to gain any marks. Of those 40% the majority just substituted the maximum force and extension into the equation from the formula sheet. Nearly half of the students that did determine the area did not do it accurately enough to gain full marks.

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In question 03.3 75% of students were able to explain that stiffness increases as compression increases but only 40% of students explained that the graph showed this because the ratio (or gradient) increased.

QUESTION 04

Question 04.1 required students to name the particles created when a free neutron decays. When the command word is name, students must name the particles rather than providing the symbols. 50% of students named all three particles correctly. Of those that didn't, some missed the antineutrino or incorrectly named the neutrino.

Question 04.2 required students to describe what happens when a free neutron collides with an antineutron. 50% of students gained two marks for correctly stating that annihilation occurs and that photons are released. 30% of students only gained one mark, mostly for just stating that annihilation occurs and not that photons are released.

QUESTION 05

Question 5 was a calculation to determine if a collision was elastic or inelastic. The question was intended to be relatively straightforward as it was not contextual, and all the information was clearly presented with a diagram.

Only 10% of students failed to score any marks, by not recognising that they had to use the conservation of momentum to determine the unknown velocity. 90% of students did gain 2 marks for determining the unknown velocity, and the vast majority of those went on to correctly calculate the kinetic energy after the collision. 10% of students, who correctly calculated the kinetic energy after the collision, did not gain the first mark as there was not a clear comparison and conclusion.

QUESTION 06

This question was about a person jumping from an aircraft wearing a parachute. The context was slightly unfamiliar as there was a horizontal wind acting.

In question 06.1 only 20% of students gained full marks. The most common error was that students did not draw a continuous curve with the correct shape above and below the x-axis so could not gain either of the first 2 marks. Some students did show the line as horizontal but did not label the y-axis at this point with –16 m s^{-1}.

Question 06.2 was a straightforward question that required students to resolve vertically and use $\boldsymbol{s = vt}$ with the vertical component of $\boldsymbol{v}$. 88% of students did at least one of the steps correctly, and 70% of students did both and gained both marks.

Question 06.3 was quite challenging to fully answer with only approximately 20% of students gaining all four marks. About 15% of students fully understood how to answer the question but only gained 3 marks because they either did not draw the tangent accurately, or the tangent line was too short. Approximately 40% of students got started by drawing a tangent and determining the acceleration, but were unable to use Newton's second law correctly.

Question 06.4 required students to explain the shape of the graph, but many students did not refer to the gradient of the line in their answer. Fewer than 10% of students gained full marks on the question. Approximately 75% of students gained at least one mark, with those who only gained one mark simply stating that as velocity increased, air resistance or drag

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increased. The vast majority of weaker answers did not attempt to explain how the forces acting on the person affected their acceleration.

QUESTION 07

This question was about a source of nuclear radiation. It started with a straightforward calculation, but the rest of the question parts, about the source being used in a school laboratory, were quite challenging for students.

Question 07.1 required students to calculate the specific charge. Although nearly 75% of students gained full marks, approximately 10% of students did not use the correct number of protons with the charge or the correct number of nucleons with the nucleon mass; some students got them the wrong way round.

In question 07.2 only 30% of students gave a correct reason as to why the half-life meant that it is suitable for use in school laboratories. Those that did answer correctly mainly used the idea that the activity would not drop to the level where it needed to be replaced. Some students answered with the idea that the activity would not change much over several years, and this was given the mark.

Question 07.3 was answered better than 07.2 with 45% of students correctly explaining that the alpha particle collided with air particles. Several incorrect answers referred to air resistance, and many did not answer in terms of collisions.

Only 25% of students gained 2 marks on question 07.4; the majority of responses lacked the detail required. 30% of students gained one mark for stating that the teacher should subtract the background count rate from the reading, but failed to describe how the background count should be measured.

Question 07.5 was very challenging with only 15% of students gaining a mark. Many students stated that the radium-225 was emitting beta and gamma radiations despite being told in the question that it only emits alpha particles. Most students answered in terms of the background count; despite being told it was the corrected count rate in the question and having just answered an earlier question about the corrected count rate.

QUESTION 08

The question started with three calculations for a disc that slides down a ramp and then across some ice. They then had a challenging question about measurements.

Question 08.1 required students to use the density equation to determine the diameter of the disc. The question was well answered with nearly 75% of students gaining all three marks. Nearly all students were able to use the density equation correctly, but a small number used the wrong formula for the volume of a cylinder, and some students did not double the radius to determine the diameter.

In question 08.2 most students used an equation of motion to determine the acceleration; only a small number used the conservation of energy to answer the question. 75% of students got started, mostly by calculating the acceleration down the slope and scored at least one mark. Some students then did not use the component of the weight down the slope and the friction, so were unable to gain further marks. 55% of students did gain full marks for calculating the angle correctly.

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Question 08.3 was relatively straightforward with 80% of students gaining both marks. Incorrect answers mainly calculated the time it would take to travel the distance at a constant speed.

Question 08.4 was very challenging for students to gain 2 marks. Explaining why the measurement was done in different locations proved difficult for students. 50% of students were able to score one mark for explaining that the measurement was repeated in the same location to minimise the effect of random errors.

QUESTION 09

Students found question 09.1 very demanding. Only 10% of students gained a mark on this question. Many students mentioned reaction time, but most did not answer in terms of variation in reaction time. Most students did not use the data in the table to answer the question, so very few explained that the values for t were too short to measure with a stopwatch without getting a large (percentage) uncertainty in the measurement.

In question 09.2 slightly over 75% of students were able to suggest one way the data could be improved; most suggested either using a greater range of data or repeating and taking a mean. Of the 35% that gained two marks, most suggested both.

In question 09.3 the majority of students wrote vague answers that were not creditworthy. Answers like ‘record the data to more significant figures’ were not marked correct as students should realise that processed data should have the same number of significant figures as the measured values.

Only 1% of students gained all three marks on question 09.4. This tiny minority were the only ones to compare the equation of motion to the equation for a straight line. 65% of students described a suitable straight-line graph to draw, but only 35% explicitly stated how $\boldsymbol{g}$ can be calculated from the gradient.

QUESTION 10

The majority of students gained at least 1 mark on question 10.1. 90% of students were able to do one of the steps in the calculation and gain one mark. 80% of students did two correct steps and gained two marks. Some students did not use the radius to determine the cross-sectional area or had a power of ten error.

Question 10.2 required students to use trigonometry to determine the angle and then use that angle to determine a force. Approximately 50% of students determined the angle correctly and 35% used that to correctly calculate the force. Of the students who did not score any marks, most got confused as to which lengths to use to determine the angle.

Question 10.3 was more challenging than the two calculations that went before it. 40% of students scored one mark. The most common correct answer was that force or tension decreases as extension or strain decreases. 20% got two marks but only 3% gained all three marks.

QUESTIONS 11-24

Generally, the multiple-choice questions were very well answered, with questions 13, 16 and 17 being the most accessible. Questions 14, 15 and 24 were the most demanding.

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